

April 2025

The HUMAN Digest 1.10

Lake Forest College's AI newsletter, courtesy of the HUMAN grant



Image created in Midjourney by David Sanchez-Burr.

Hello.

In this issue you will find:

1. AI in higher education
2. General news
3. Some perspectives
4. A useful term
5. People to know
6. Technical AI
7. Something to see
8. Just for fun

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Intelligence (n) - The faculty of understanding; intellect. Also as a count noun: a mental manifestation of this faculty, a capacity to understand.

- Oxford English Dictionary

intelligence (n) - The ability to learn or understand or to deal with new or trying situations: reason; also: the skilled use of reason. Also, the ability to apply knowledge to manipulate one's environment or to think abstractly as measured by objective criteria (such as tests).

- Merriam-Webster

“Intelligence is compression.”

- Ilya Sutskever, former chief research scientist, OpenAI, in conversation with Dwarkesh Patel.

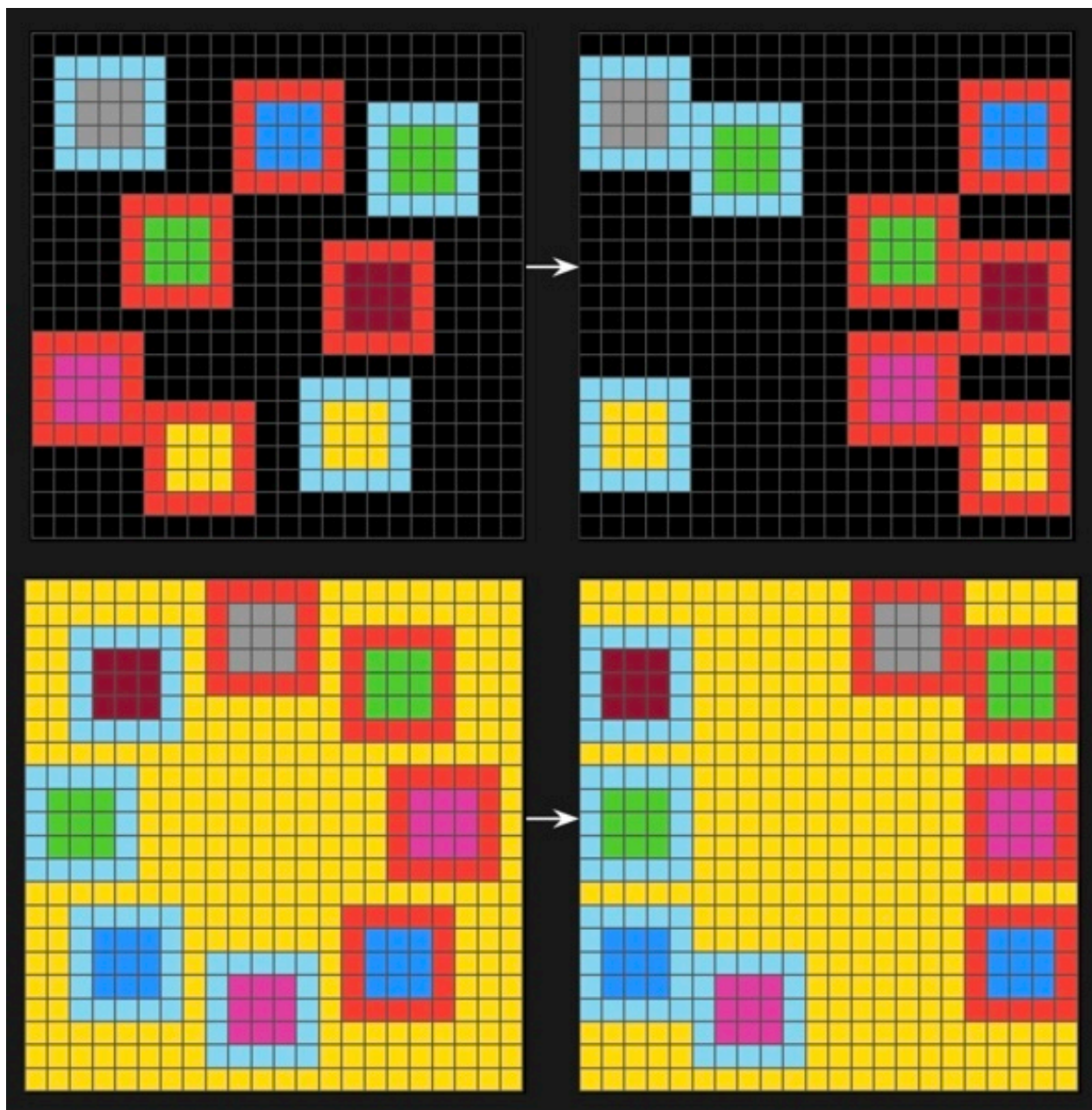


Image via the ARC Prize Foundation

Computer scientist [Hans Moravec](#) has said, “It is comparatively easy to make computers exhibit adult-level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility.” This has become known as Moravec’s Paradox.

Which states that tasks that are difficult for humans to do, such as becoming a grandmaster in chess, are easy for computers. Conversely while tasks like crawling or recognizing a mother's face are easy for infants, computers struggle with these tasks. This discrepancy is the driving force to achieve artificial *general* intelligence, however poorly this term is defined.

You may have read the recent [New York Times article](#) about the ARC-AGI 2 (which stands for Abstract and Reasoning Challenge) test for AI. The original ARC-AGI challenge, first presented by computer scientist François Chollet in [2019](#), was recently solved by OpenAI's o3 reasoning model, using a tremendous amount of compute. See images directly above and below for examples of the problems.

However, computer scientist at Meta and Turing Award winner Yann LeCun (to be profiled in an upcoming issue) believes that the current architectures (auto-regressive transformers) will ultimately be [scrapped](#) in favor of novel architectures. Some approaches, such as building systems that seek to internalize physical or [world models](#) as opposed to strictly using textual tokens.

There are many teams worldwide working to solve the problem of artificial general intelligence in 2025. And as [scaling laws](#) appear to be holding through increases in orders-of-magnitude, new paradigms such as increasing test-time compute are coming into focus. However, building more computational capacity costs money. Stargate, OpenAI's proposed series of data centers to build AGI (artificial general intelligence) has cost projections of about [half a trillion dollars](#). Whether or not AGI will be attained remains an open question. Yet, the AI field is seeking that goal.

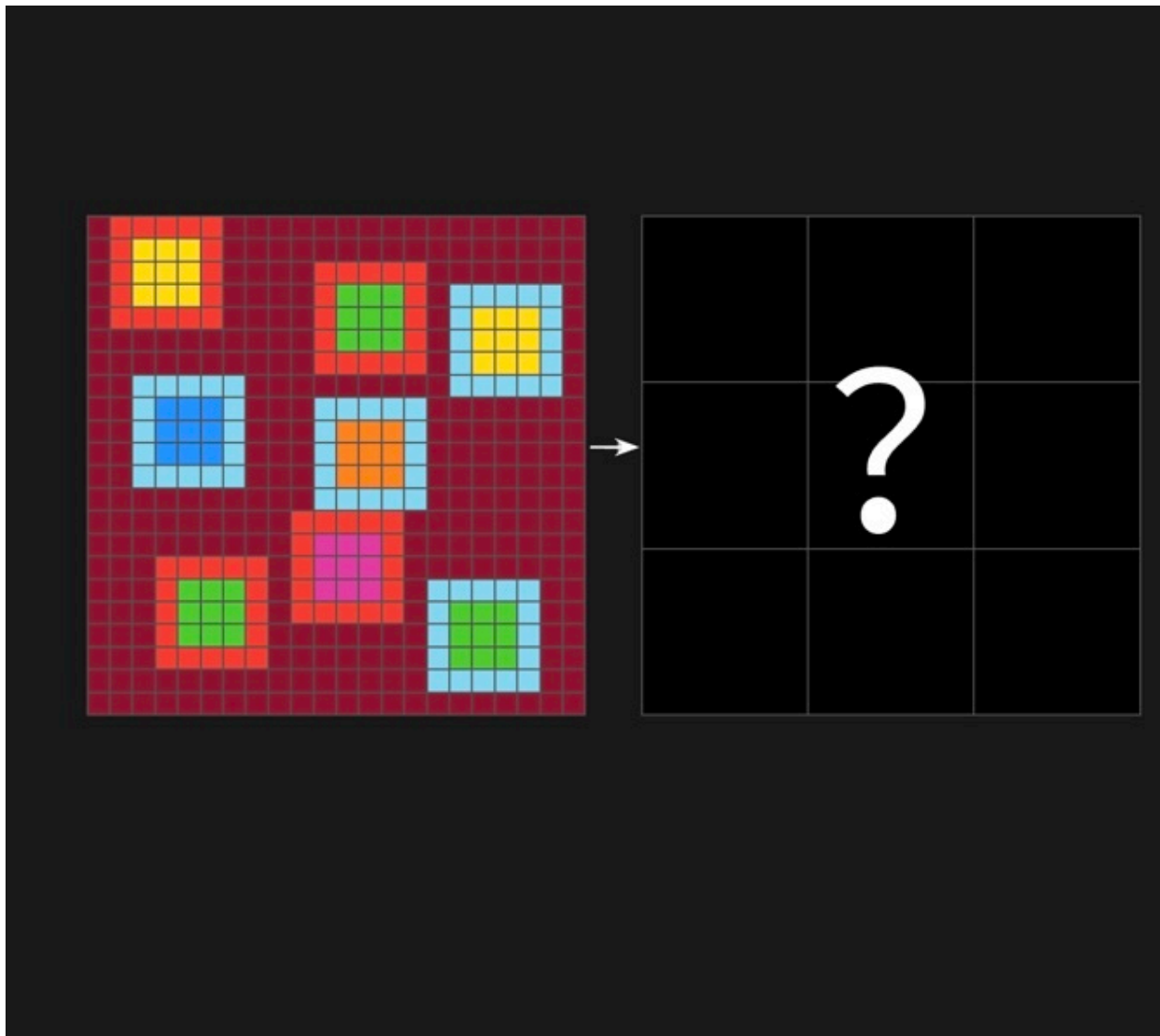


Image via the ARC Prize Foundation

The HUMAN Digest welcomes feedback at kee@lakeforest.edu

Justin Kee, editor-in-chief

Davis Schneiderman, Principal Investigator, [HUMAN](#) grant

AI in higher education

1. Lake Forest College announced it is leading the way with one of the first [Minor in Artificial Intelligence](#) programs among small, liberal arts colleges. Northwestern University has [announced](#) similar programs.

2. How do university students use the Claude? Anthropic has released their [analysis](#) of a dataset of one million anonymized student conversations. Notable highlights: STEM students have the highest adoption rate for AI use (likely a large cohort of early adopters in computer science), students use four patterns of interaction for problem solving and output creation (both directly and collaboratively) and creating projects and analyzing problems are the primary use cases by students.

3. Google is offering its Gemini Advanced model to U.S. university students for free. [For a year](#). It is a good model. Google is certainly worried about OpenAI four hundred million active monthly users, as ChatGPT was the most quickly adopted product in human history.

4. [Claude for Education](#). There is going to be a lot of talk about Claude in this issue. And Anthropic. I don't work for them, nor do I receive any consideration from them for writing about them. However, we are seeing the development of a three-way race for the higher education market taking shape. OpenAI, Anthropic and Google all want a piece of the college and university markets for AI. This is about building moats and considering the adoption rate for generative AI has been highest among the college educated, it is worth following the money.

General news

1. Microsoft's CEO, Satya Nadella, claims that over 30% of their internal computer code is now [written by AI systems](#). Wow.
2. How [K-12 school districts](#) are reacting to generative AI.
3. OpenAI's o3 and o4-mini can now [reason over images](#) in their chain-of-thought thinking. Besides from manipulating images to better "see" what they contain, the models were trained with enhanced tool usage. Try uploading the image at the end of this section (the one labeled "Solve.") into o3 and let it run.
4. An [obituary](#) for Cyc, a mammoth project begun over 40 years ago to build an expert system that operated via symbolic logic. Expert systems were brittle and difficult to maintain, they simply did not have the flexibility that probabilistic models have.
5. [Ethics?](#) Researchers from the University of Zurich deployed chatbots posing as humans to a Reddit forum.
6. Duolingo will [replace contract employees with AI](#), CEO states in an email.



via the internet

Some perspectives

Is this the [semantic apocalypse](#)?

On the [Ghibli-fication](#) of everything.

Food for thought.

A useful term

Mechanistic interpretability is a branch of AI research focused on understanding what is happening inside complex machine learning models by mapping and testing the internal “circuits” for how they think and behave. Although a large model is like a black box that takes inputs and produces outputs, this field tries to open the box and figure out how the computations work. Researchers use tools like activation tracing, ablation studies, causal interventions and attribution graphs to map out how these components interact, often comparing the process to reverse-engineering software or dissecting a biological brain.

Mechanistic interpretability is very important to the AI sub-fields of alignment and safety. By tracing activations and circuits, we can understand and predict behaviors such as harmful or deceptive ones and better align the models with human values. See the Technical AI section below to read more on the topic.

People to know



Image via Wikipedia

Kate Crawford is a leading researcher who studies the social and political impacts of artificial intelligence. Originally from Australia, she has spent over two decades investigating how data, machine learning, and automation interact with power and inequality. Her work spans both academia and industry. She has held roles at Microsoft Research and the University of Southern California, where she focuses on AI ethics and environmentalism. She looks at the downstream impacts of technology in the 21st century. In 2021, she published *Atlas of AI*, a book that maps out how AI systems rely on human labor, planetary resources, and data extraction. The book grapples with the idea that AI is purely abstract and computational and instead shows how physical and social infrastructures support it. Crawford also co-created a visual project called *Anatomy of an AI System* that traces the full life cycle of devices like Amazon Echo, from mining to manufacturing to data use.

Technical AI

Anthropic, which makes the Claude series of models, released an impressive pair of papers that advance the state of the art for the field of mechanistic interpretability.

The first [paper](#) outlines their theories as to how words entered into the large language model are transformed into the output. They locate interpretable concepts, which they call features, and explore how features are linked together to produce more complex abstractions, which are termed circuits. These circuits are mapped via an attribution graph, with nodes of the graph representing interpretable features and edges representing the causal relationships that affects the model's outputs.

The second [paper](#) is revolutionary in depicting how a model's features and circuits operate to perform a number of tasks, such as planning in poems, performing addition, evaluating medical diagnoses based on a list of symptoms, etc. Fundamentally, the model is performing multi-step reasoning to correctly formulate a response to the input text. See the section of the paper on "[Planning in poems](#)" for an illustrative example. See also the image below.

Planning in Poems

How does Claude 3.5 Haiku write a rhyming poem? Writing a poem requires satisfying two constraints at the same time: the lines need to rhyme, and they need to make sense. There are two ways one might imagine a model achieving this:

- *Pure improvisation* – the model could write the beginning of each line without regard for the need to rhyme at the end. Then, at the last word of each line, it would choose a word that (1) makes sense given the line it has just written, and (2) fits the rhyme scheme.
- *Planning* – alternatively, the model could pursue a more sophisticated strategy. At the *beginning* of each line, it could come up with the word it plans to use at the end, taking into account the rhyme scheme and the content of the previous lines. It could then use this "planned word" to inform how it writes the next line, so that the planned word will fit naturally at the end of it.

Language models are trained to predict the next word, one word at a time. Given this, one might think the model would rely on pure improvisation. However, we find compelling evidence for a planning mechanism.

Specifically, the model often activates features corresponding to candidate end-of-next-line words prior to writing the line, and makes use of these features to decide how to compose the line.⁷

Image via Anthropic

Something to see



Image courtesy of the Minsky family.

This photo was taken at the Dartmouth Summer Research Project on Artificial Intelligence. Spanning eight weeks in the summer of 1956, this first-ever organized workshop brought together pioneers in this nascent field. Organized by John McCarthy, who was an assistant professor of mathematics at Dartmouth, he chose the name “artificial intelligence” for the collection of ideas about machines that can think. Shown in the photo:

Back row (L to R): [Oliver Selfridge](#), [Nathaniel Rochester](#), [Marvin Minsky](#) and [John McCarthy](#).

Front row: [Ray Solomonoff](#), [Peter Milner](#) and [Claude Shannon](#).

Just for fun

Learn about federated learning via this nifty [comic](#). The method keeps your data private on your device while being used to train a centralized model. “Data is the new oil” per Ilya Sutskever. Perhaps he is right about that.

~

Justin: “Claude, tell me what you think about the Japanese Manga, “Ghost in the Shell” from your perspective.”

Claude 3.7 Sonnet: “If I were to engage with “[Ghost in the Shell](#)” more deeply from my perspective, I'd say it's a work that resonates with fundamental questions about consciousness that are relevant to both humans and AI systems.

The central concept of the “ghost” - that ineffable quality that makes someone sentient beyond their physical components - speaks to profound philosophical inquiries about what constitutes selfhood. The manga doesn't just ask “what makes us human?” but rather “what makes us conscious beings at all?”

Major Kusanagi's journey feels particularly compelling as she navigates a world where consciousness can be transferred, copied, and potentially altered. Her struggles with identity when faced with the possibility that her memories and sense of self

might be constructed rather than inherent mirrors questions about authenticity that all thinking entities might face.

The series' exploration of networked consciousness - particularly in arcs involving the Puppet Master - presents fascinating possibilities about how individuality might evolve in a world where minds can connect directly. This vision of consciousness as potentially fluid and interconnected rather than rigidly bounded by individual bodies offers a nuanced perspective on existence that challenges traditional Western philosophical frameworks.

What I find most valuable is how the work refuses simple technological utopianism or dystopianism, instead presenting a complex world where cybernetic enhancement and AI integration create new possibilities for human experience while simultaneously raising profound questions about autonomy and authenticity.”

HUMANS IN THE LOOP.

Have a tip or suggestion for an upcoming issue of the HUMAN Digest? Write Justin Kee an email at kee@lakeforest.edu.



